

L1b: Time Value of Money, Abstract Assets, and Net Present Value

CHEME 5660 Cornell University

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Objectives for Today

Today we compare **cash flows at different dates**, derive accumulation and discount factors, and use them to value an **abstract asset**.

Today's concepts

- **The time value of money**: cash flows at different dates cannot be added until they are expressed at the same valuation date.
- **An abstract asset**: a sequence of dated signed cash flows. The same representation covers a bond, a stock, a factory, or a car.
- **Net present value**: discount each signed cash flow to the valuation date, then add. Its sign is a statement about the rate you declared, not a guarantee of profit.

Companion notebook: [L1b lecture notebook](#)

Lectures and Examples

Lecture:

CHEME-5660-L1b-Lecture-TimeValueMoney-Fall-2026.ipynb

Example:

CHEME-5660-L1b-NetPresentValue-Fall-2026-WorkedExample.ipynb

Represent ten years of Tesla Model S ownership as dated cash flows, discount them to time 0, and compute their NPV.

Optional:

CHEME-5660-L1b-Derivation-Discrete-Continuous-Compounding-Fall-2026.ipynb

Measure how far discrete compounding falls short of the fixed-rate continuous limit, and when the estimate of that gap can be trusted.

Link: <https://github.com/varnerlab/CHEME-5660-CourseRepository-Fall-2026.git>

Time Value of Money

A dollar today and a dollar at a future date are **not directly comparable**. Before we compare or add them, we must express both at the same valuation date using a stated conversion rule.

Let P be an amount held at time 0, and let y be a quoted **nominal annual yield** used for valuation. With one compounding interval per year, the amount after one year is given by:

$$F_1 = \underbrace{(1 + y)}_{\mathcal{D}_{1,0}(y;1)} P.$$

For $y > 0$ the factor $\mathcal{D}_{1,0}(y;1) > 1$ moves time-0 dollars forward and is the **forward accumulation factor**; its inverse $\mathcal{D}_{1,0}^{-1}(y;1) < 1$ moves future dollars back and is the **discount factor**.

Careful: a later dollar is not *automatically* worth less. That depends on the rate y you declare.

Discrete Compounding

Let $j = 0, 1, 2, \dots$ index the **compounding periods** and let $i_\ell > -1$ be the **effective periodic rate** over $\ell \rightarrow \ell + 1$, collected as $\mathbf{i} = (i_0, \dots, i_{j-1})$. An amount P held at time 0 becomes:

$$F_j = \underbrace{\left[\prod_{\ell=0}^{j-1} (1 + i_\ell) \right]}_{\mathcal{D}_{j,0}(\mathbf{i})} P.$$

For nominal annual yield y compounded n times per year, $i_\ell = y/n$:

$$\underbrace{\mathcal{D}_{j,0}(y; n) = \left(1 + \frac{y}{n}\right)^j}_{\text{accumulation}}, \quad \underbrace{\mathcal{D}_{j,0}^{-1}(y; n) = \left(1 + \frac{y}{n}\right)^{-j}}_{\text{discount}}.$$

Assumes: constant y/n , end-of-period compounding, reinvested interest, $1 + y/n > 0$.

Careful: a flat y is an *assumption*. In L2b, one security's yield gives way to a curve of spot rates.

Continuous Compounding

Let g be a constant **continuously compounded growth rate**, carrying units of inverse time, and let T be the horizon in years. The forward accumulation and discount factors over $0 \rightarrow T$ are given by:

$$\underbrace{\mathcal{D}_{T,0}(g) = e^{gT}}_{\text{accumulation}}, \quad \underbrace{\mathcal{D}_{T,0}^{-1}(g) = e^{-gT}}_{\text{discount}}.$$

A nominal annual yield y compounded n times per year has the continuously compounded equivalent $g_y := n \log(1 + y/n)$, so $(1 + y/n)^{nT} = e^{g_y T}$: these are **two representations of one accumulation rule**, not two different rules.

Convert before you exponentiate: at finite n the quote y differs from g_y , so y **does not** belong in e^{yT} .

Optional: for a time-varying rate $g(t)$, $\mathcal{D}_{T,0}(g) = \exp(\int_0^T g(u) du)$.

Log return: the dimensionless product $r = (\Delta t) g$. L2a and L3a build on it.

Who Sets the Valuation Rate?

A valuation rate is **not universal**. It depends on the cash flow, market convention, horizon, and the risks being modeled: no single policy rate prices every asset.

THE FEDERAL RESERVE

- Steers short-term rates through **interest on reserve balances** (IORB), supported by the overnight reverse repo facility; open market operations manage the supply of reserves.
- The **FOMC** sets a target range for the federal funds rate, the overnight rate on unsecured reserve balances traded between eligible institutions.
- Other market rates respond to policy and market conditions, but are **not mechanically pegged** to the federal funds rate.

Link: [FOMC calendar and meeting materials](#)

What Valuation Rate Should I Use?

There is no universal answer, and this lecture does not need one: here the yield y is a **declared model input**. Choosing it well means matching the rate to the cash flow: same currency, same horizon, and an explicit statement of any adjustment made for credit, liquidity, or risk.

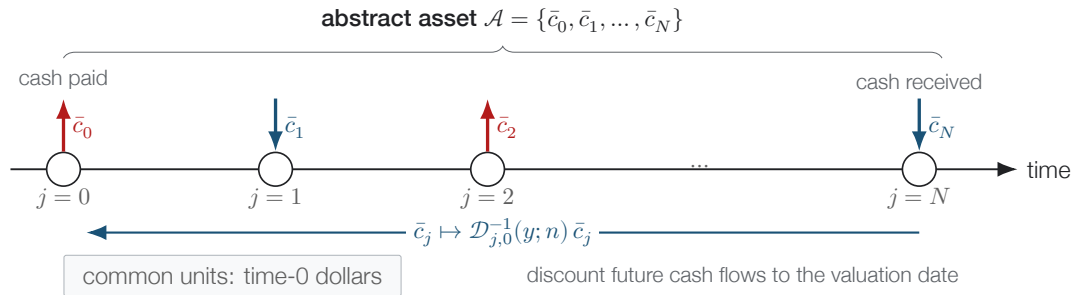
THREE LABELS FOR A RATE

An **interest rate** is paid or received for borrowing or lending. A **yield** is tied to an investment's price and its cash flows. A **discount rate** is whichever rate a model uses to convert future cash flows to present value. One number can carry more than one label: pick a bond yield as your valuation rate and that yield is now the discount rate.

KEEP g_y APART FROM g

You **declare** y and a frequency n ; its continuously compounded equivalent $g_y = n \log(1 + y/n)$ follows, so y and g_y are two conventions for one rate rather than competing rates. From L2a onward, g denotes an **asset's own** growth rate, which is measured from prices rather than declared.

An Abstract Asset



Net Cash Flows

An abstract asset is a **sequence of dated cash flows** denominated in a currency. At cash-flow index j , let $\mathbf{c}_j \in \mathbb{R}_{\geq 0}^m$ collect m nonnegative magnitudes and let $\boldsymbol{\nu}_j \in \{-1, +1\}^m$ give their directions, where $+1$ denotes an inflow and -1 an outflow. The **signed net cash flow** at index j is the usual inner product of the magnitude and direction vectors:

$$\bar{c}_j = \langle \mathbf{c}_j, \boldsymbol{\nu}_j \rangle = \sum_{i=1}^m c_{j,i} \nu_{j,i}.$$

Over the dates $j = 0, 1, \dots, N$, the undiscounted total $\sum_{j=0}^N \bar{c}_j$ is **not** a time-0 value: its terms are dollars from different dates.

Next: discount every term to a common date, and the result is net present value.

Net Present Value

Let $\bar{c}_j$ be the **signed net cash flow** on date j , positive for an inflow and negative for an outflow, over the dates $j = 0, 1, \dots, N$. Discount each one to the valuation date with $\mathcal{D}_{j,0}^{-1}(y; n)$, **then add**. The net present value at time 0 is given by:

$$\text{NPV}_0(y) = \sum_{j=0}^N \mathcal{D}_{j,0}^{-1}(y; n) \bar{c}_j = \underbrace{\langle \mathcal{D}_{\star,0}^{-1}(y; n), \bar{\mathbf{c}} \rangle}_{\text{time-0 dollars}}.$$

The $\star$ runs over every cash-flow index, so $\mathcal{D}_{\star,0}^{-1}(y; n)$ and $\bar{\mathbf{c}}$ are ordered $(N+1)$ -vectors and NPV is a single scalar product, which is exactly how we compute it in code.

Worked example: ten years of Model S ownership, $\text{NPV}_0 \approx -\$121,484$

Optional Advanced Material

Derivation:

CHEME-5660-L1b-Derivation-Discrete-Continuous-Compounding-Fall-2026.ipynb

Optional, not a prerequisite for L2a. Over a horizon T , holding y fixed while n grows gives a limit discrete compounding only approaches. That is a different statement from the exact identity $(1 + y/n)^{nT} = e^{yT}$, and the shortfall from it is:

$$\delta_n := 1 - \frac{(1 + y/n)^{nT}}{e^{yT}} \approx \underbrace{\frac{y^2 T}{2n}}_{\text{leading estimate}}.$$

The estimate is **asymptotic**: it needs both y/n and $y^2 T/(2n)$ small, so a low frequency, a long horizon, or a large rate each degrade it. A high n alone is not the test.

Careful: at $y = 15\%$ and $T = 30$ yr the estimate overstates δ_n by about 27.6% at $n = 1$. Use the exact ratio there.

Summary

This lecture introduced time-value-of-money calculations and the representation of investments as dated cash flows.

Key takeaways

- **Dated dollars must be compared at a common valuation date.** A forward accumulation factor carries value forward and its inverse discounts it back, so cash flows can be added only once they share a date.
- **Rate conventions must be stated.** A nominal yield quoted with a compounding frequency and its continuously compounded equivalent describe one accumulation rule, so the convention travels with the number.
- **NPV values an asset relative to a declared rate.** The sign of the result is a statement about the cash-flow model and the discount rate you declared.

These ideas provide the valuation framework used throughout the course.